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Large human mobility datasets remain spatially unrepresentative across geographic scales

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GPS-based human mobility data provide unique opportunities to study population activity, exposure, and access at fine geographic scales. However, large samples do not necessarily provide representative samples. existing evaluations have largely focused on whether device counts reproduce population totals across geographic areas; this captures only one aspect of representativeness. We distinguish two dimensions of representativeness: *population representation*, whether inferred home counts are allocated across communities in proportion to census populations, and *spatial representation*, whether inferred homes reproduce the expected distribution of residents within communities. We evaluated these dimensions across five commercial mobility products in two U.S. states (Florida and California) at county, census tract, and census block group scales. Population representation was benchmarked against the American Community Survey estimates, while spatial representation was evaluated for three products with device level home locations by comparing observed home-location spacing with reference patterns simulated within residential land and weighted by independent population-density estimates. Our evaluation results suggest that, population mismatch consistently increased at finer geographic scales, whereas spatial mismatch showed no consistent scale dependence. The two dimensions were also weakly related: 44.0% to 54.8% of the evaluated population lived in areas classified as representative on only one dimension. Moreover, we found that increasing sample size primarily reduced random variation in population allocation but did not correct census population mismatch, while retaining more inferred homes did not consistently improve spatial representation. Demographic and built-environment correlates also differed between the two outcomes. These findings show that sample size and agreement with aggregate population counts alone are insufficient to establish spatial representativeness. Evaluations of GPS-based mobility data should therefore consider both dimensions at the geographic scale of the intended analysis.

human mobility | spatial bias | population representation | spatial representation | geographic scale

Significance

GPS-based mobility data inform studies of environmental exposure, service access, and disaster response. We distinguish two dimensions of representativeness: whether communities are sampled in proportion to their populations and whether inferred homes follow expected residential patterns within communities. Evaluating five commercial datasets in Florida and California, we found that these dimensions were weakly related. Larger samples reduced random variation in population allocation without eliminating systematic mismatch and did not consistently improve spatial representation. Thus, agreement with population counts can mask spatial bias in data used to compare communities. Both dimensions should be evaluated at the geographic scale of the intended analysis before mobility data inform research and policy.

Introduction

Large-scale human mobility data generated from smart mobile devices such as smartphones, watches, and other devices have transformed how researchers observe population movement. Their large sample sizes and fine spatial and temporal resolution enable applications including dynamic population mapping, analyses of urban interaction, pandemic response, and the measurement of environmental inequality (1–5). Many of these applications rely on inferred home locations to connect sampled devices with residential populations, environmental exposures, and access to services. As mobility data are increasingly used to study population patterns at fine geographic scales, understanding how well these samples represent the underlying population becomes increasingly important.

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S.J. and X.Z. designed the study. S.J. and X.Z. collected the data. S.J. and A.R. performed the analysis and interpreted the results. A.R. and S.J. wrote the first draft. All authors reviewed and approved the manuscript.

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Large sample size, however, does not necessarily imply representativeness. Commercial mobility products reflect multiple stages of selection and processing. Individuals differ in smartphone access, application use, and willingness to share location, while providers collect data through selected applications and apply their own privacy, quality control, and aggregation procedures (6). The resulting location traces can contain systematic gaps and positional uncertainty (7), and additional uncertainty is introduced when incomplete or irregular traces must be converted into a stable residential locations (8–10). Aggregate and synthetic products are further shaped by provider definitions, processing rules, and calibration targets. Because these processes may affect population groups and geographic areas differently (11, 12), increasing the number of observations may reduce random sampling variation without correcting systematic differences between the observed sample and the population it is intended to represent, reflecting the broader big data paradox (13, 14).

Recent studies have primarily evaluated representativeness by comparing user counts or inferred resident counts with census or administrative population benchmarks (12, 15–17). These studies have identified differences in coverage across demographic and socioeconomic groups, levels of urbanization, time periods, and data providers. While such comparisons provide important information about whether sampled residents are distributed across geographic areas in proportion to the underlying population, they provide limited information about how inferred residents are spatially distributed *within* those areas.

This distinction suggests two related but different dimensions of representativeness. We define *population representation* as whether inferred residents are allocated *among* geographic areas in proportion to the underlying population, and *spatial representation* as whether inferred home locations reproduce expected residential patterns *within* those areas. A census tract, for example, may contain approximately the expected number of sampled residents while those inferred homes are concentrated in only a small portion of its residential area. Conversely, an area may contain a disproportionate share of sampled residents while their locations still follow the expected local residential pattern. Agreement with population counts therefore does not necessarily imply that the spatial distribution of inferred homes is representative.

Geographic scale further complicates these relationships. Aggregation can conceal local population mismatch when overrepresentation in one neighborhood offsets underrepresentation in another, consistent with the scale component of the modifiable areal unit problem (18). Prior work has shown that agreement between mobile device counts and census population often weakens at finer geographic scales (16). However, this evidence is largely based on count comparisons drawn from a limited set of data sources; it remains unclear whether this pattern persists across different commercial mobility products or whether spatial representation changes similarly with geographic scale. This distinction is particularly important for studies of neighborhood-level accessibility, exposure, equity, and travel behavior, where county-level agreement may not reflect data quality at the census tract or block group level (19).

Sample size presents a separate but closely related challenge. Areas with larger mobility samples may also differ in population density, demographic composition, urban form, provider coverage, and trace quality. A cross-sectional relationship between sampling rate and representation therefore cannot establish whether increasing the number of observations itself improves representativeness. For population representation, larger samples could approach the census distribution, or simply reproduce the geographic distribution already present in the product with less random variation. For spatial representation, retaining more inferred homes may similarly reduce random variation without improving agreement with expected residential patterns. Controlled sampling experiments are therefore needed to distinguish improvements caused by reduced finite-sample variation from systematic mismatch that persists as sample size increases (14).

To address these gaps, we define spatial bias as a systematic mismatch between the observed sample and the real population, then evaluate it using two components: population and spatial representation. *Population representation* evaluates allocation among geographic units by comparing each area's share of estimated residents with its share of the American Community Survey population. The population representation ratio records the direction of mismatch, while the population representation score summarizes its magnitude on a common scale. Total variation distance, Jensen Shannon divergence, and the population weighted Gini coefficient provide complementary summaries of statewide population mismatch. *Spatial representation*, by contrast, evaluates the internal spacing of inferred homes conditional on the number observed. For each area, we generate reference point patterns with the same number of homes within mapped residential land, with sampling probabilities proportional to an independent estimate of residential population density. The spatial representation score then compares the observed mean five nearest neighbor distance with the simulated reference. This measure evaluates the distribution-level spatial configuration of inferred homes rather than the accuracy of individual home coordinates. Thus, population representation evaluates allocation *among* areas, whereas spatial representation evaluates spatial configuration *within* an area.

We apply this framework to five mobility products in Florida and California: Gravy, Anon_Vendor, Unacast, Advan, and Replica, across seven provider-state combinations three geographic scales: county, census tract, and census block group. Population representation is assessed for all seven combinations. Device level inferred homes are available for Gravy, Anon_Vendor, and Unacast; because these three products are processed with the same home inference method, they are also evaluated for spatial representation. We use spatial statistics and distributional measures to compare products and scales, joint score analyses test whether population and spatial representation identify the same areas, multinomial resampling and controlled thinning examine the role of sample quantity, and spatial error models and random-effects synthesis estimate associations with demographic, socioeconomic, and built-environment factors while accounting for residual spatial dependence.

The study addresses four research questions:

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1. How do population and spatial representation vary across mobility products and geographic scales, and how spatially structured is population mismatch?

2. To what extent do population and spatial representation identify the same geographic areas as adequately represented?

3. Does increasing effective sample size reduce population allocation mismatch or improve the within area spacing of inferred homes?

4. Which demographic, socioeconomic, and built environment characteristics are associated with each form of mismatch after spatial dependence is considered?

This study makes four contributions. First, it distinguishes population allocation across geographic units from the spatial configuration of inferred homes within them. Second, it provides complementary reference based measures that allow aggregate and device level products to enter a common audit without treating their information as equivalent. Third, it combines a multiple product and multiple scale comparison with controlled sampling experiments that separate finite sample variation from persistent mismatch. Fourth, it evaluates local correlates of both dimensions while accounting for residual spatial dependence. The results show that (1) population mismatch increases at finer geographic scales, while spatial representation does not follow the same scale pattern; (2) the two dimensions show little agreement; and (3) increasing sample quantity does not consistently improve either outcome. Representation should therefore be evaluated for the specific product, geographic scale, representation dimension, and intended application.

Results

Population representation bias increases at finer geographic scales. Population allocation mismatch was consistently larger at finer geographic scales in every provider and state combination. Figure 1a maps the directional population representation ratio (RR_i) at the census tract scale, where an RR_i below one indicates that a tract's share of residents inferred from mobility data is smaller than its share of the census population, and an RR_i above one indicates the reverse. Underrepresentation and overrepresentation were widespread across Florida and California. Several rural and inland tracts were underrepresented, though the direction and magnitude of mismatch varied across products; this spatial heterogeneity is consistent with prior evidence that the population coverage of commercial mobility data varies across geographic scales, levels of urbanization, time periods, and data sources (16, 20). Gravy, Anon_Vendor, and Unacast each displayed interspersed clusters of underrepresentation and overrepresentation, and none maintained proportional population allocation throughout its study state. Advan showed extensive overrepresentation in California, including much of Southern California and several inland tracts. In contrast, Replica showed the closest agreement with ACS population in both states. This agreement is expected in part because Replica generates a synthetic population that is calibrated to Census controls, including ACS estimates, at small geographic scales (Replica methodology). Therefore, its agreement with ACS should not be interpreted as independent evidence of representative device sampling.

Figure 1b shows substantial variation in the spatial representation score (SRS) for the three datasets containing device level location records. The two Florida datasets had higher mean SRS values than Unacast in California. Several urban and coastal corridors contained tracts with relatively high scores, though high and low values remained spatially interspersed throughout. Unacast showed relatively low SRS across much of California, a pattern that may reflect provider specific coverage and home-inference processes, differences in residential form and land use inputs, or both.

Three distributional measures showed the same ordering across geographic scales (Fig. 1c). Total variation distance, Jensen Shannon divergence, and the population weighted Gini coefficient increased from counties to census tracts and then to census block groups in all seven provider and state combinations. This ordering is expected under geographic refinement because aggregation can offset local deviations in opposite directions (16), but the magnitude of the increase varied by product. The one sided paired Wilcoxon P value was 0.0078 for each metric and scale contrast (two-sided $P = 0.0156$). Block group total variation distance was 2.75 to 4.25 times the county value: Gravy increased from 0.079 at the county scale to 0.154 at the tract scale and 0.217 at the block group scale; Anon_Vendor increased from 0.072 to 0.221 and 0.304; and Advan in California increased from 0.075 to 0.234 and 0.315. The complete values are reported in SI Appendix, Tables S3 and S5.

The population representation score (PRS), where a score of one indicates exact proportional allocation, showed the same scale dependence for individual geographic areas. Using $PRS_i \geq 0.80$ as a descriptive criterion for adequate population representation (equivalent to $0.80 \leq RR_i \leq 1.25$), the population share living in adequately represented areas declined sharply at finer scales (Fig. 1d). For Gravy, this share declined from 72.4% at the county scale to 44.1% at the tract scale and 30.9% at the block group scale. Anon_Vendor dataset declined from 76.1% to 32.3% and 21.6%. Even Replica, which had statewide coverage near 38%, declined from 97.9% to 53.8% in Florida and from 98.6% to 56.4% in California. This ordering remained at PRS criteria from 0.60 through 0.95 (SI Appendix, Fig. S3).

Spatial representation, however, did not follow the same scale pattern. The population weighted mean SRS was 0.781, 0.812, and 0.812 for Gravy at county, tract, and block group scales, respectively. The corresponding values were 0.619, 0.693, and 0.719 for Anon_Vendor, and 0.518, 0.651, and 0.671 for Unacast (Fig. 1d). These modest numerical increases may reflect more internally homogeneous residential environments at finer scales, but paired tests across the three datasets did not support a systematic scale difference. Population weighted score distributions are shown in SI Appendix, Figs. S1 and S2. Complete SRS summaries are in SI Appendix, Tables S4 and S5.

Population mismatch was also spatially structured. Global Moran's I was positive for all 21 provider, state, and scale combinations (Fig. 1d). Nineteen of the 21 estimates were statistically significant; all tract and block group estimates had a permutation P value of

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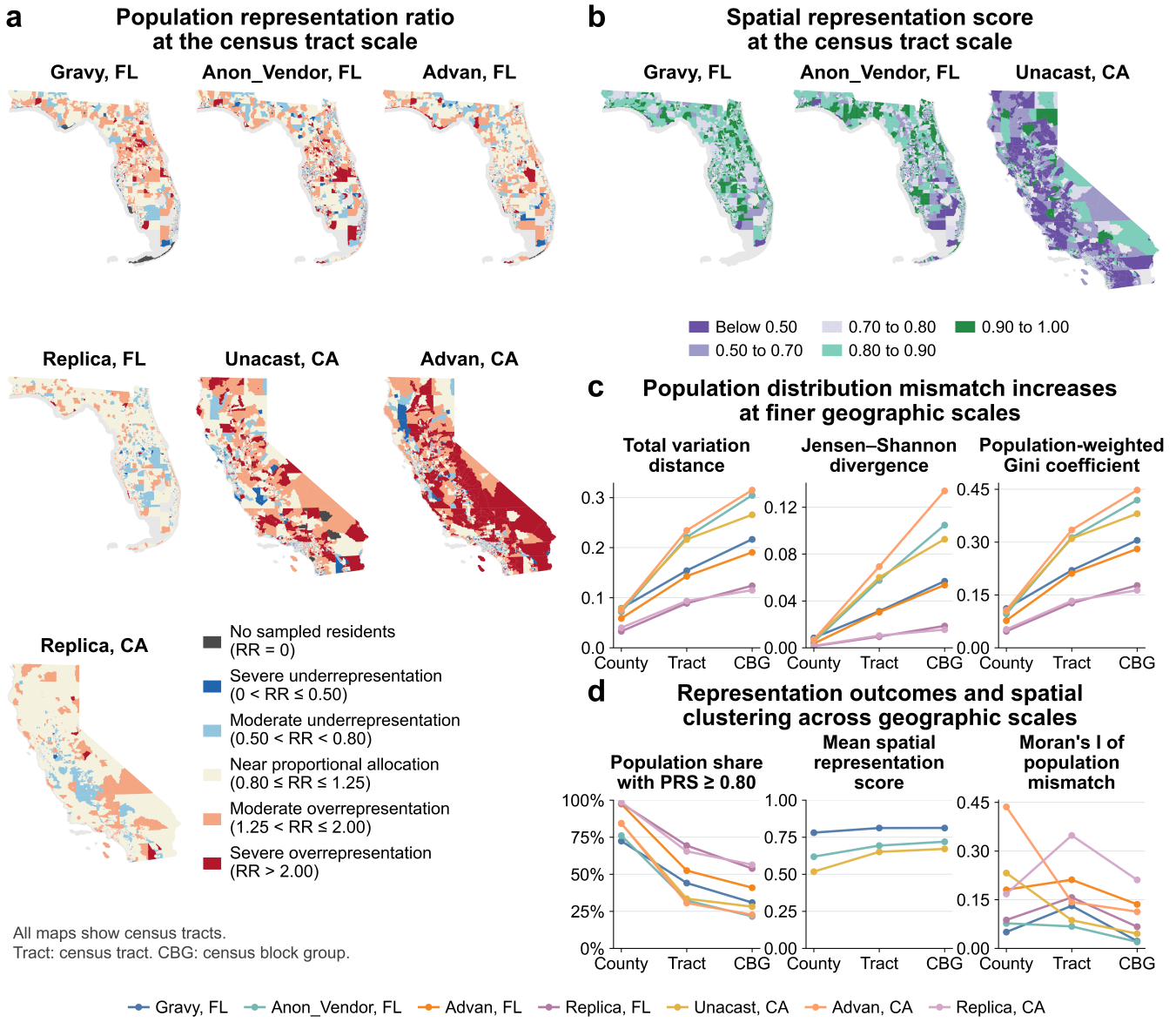


Fig. 1. Spatial bias is widespread across mobility datasets and varies with geographic scale. (a) Census tract maps show the population representation ratio (RR_i) for seven provider and state combinations. A ratio of one indicates proportional allocation relative to the American Community Survey population. The map classes distinguish no sampled residents, severe and moderate underrepresentation, near proportional allocation, and moderate and severe overrepresentation. (b) Census tract maps show the spatial representation score for Gravy, Anon_Vendor, and Unacast. Higher values indicate closer agreement between empirical home spacing and the residential reference. (c) Total variation distance, Jensen Shannon divergence, and the population weighted Gini coefficient summarize population distribution mismatch at county, census tract, and census block group scales. Lower values indicate closer agreement. (d) The first panel reports the American Community Survey population share in areas with a population representation score of at least 0.80. The second reports the population weighted mean spatial representation score. The third reports Moran's I for population representation mismatch. Spatial representation requires device level inferred home locations and is available for three datasets.

0.001, which was the minimum attainable value with 999 permutations. Moran's I did not change monotonically across scales, indicating the overall magnitude of population mismatch and its spatial concentration represent distinct properties. Results remained positive under four, eight, and 12 nearest neighbor weights and Queen contiguity. Among the 84 estimates across these four spatial weight specifications, 77 were statistically significant (SI Appendix, Fig. S4 and Table S6).

Population and spatial representation capture distinct dimensions of spatial bias. Population allocation that closely matched the census benchmark did not ensure that inferred homes followed the expected spatial pattern within an area. We examined this distinction in Gravy, Anon_Vendor, and Unacast at the tract and block group scales, where both PRS and SRS were available. Population-weighted Spearman correlations between the continuous scores were weak and negative thorughout (Fig. 2a), ranging from −0.178 to −0.010. The Gravy estimate was close to zero at the block group scale, while Anon_Vendor and Unacast showed weak negative associations at both scales. Dots in Fig. 2a show the estimated correlations, and horizontal lines show 95% confidence intervals from 500 county cluster bootstrap samples, in which counties were resampled and all tracts or block groups within the

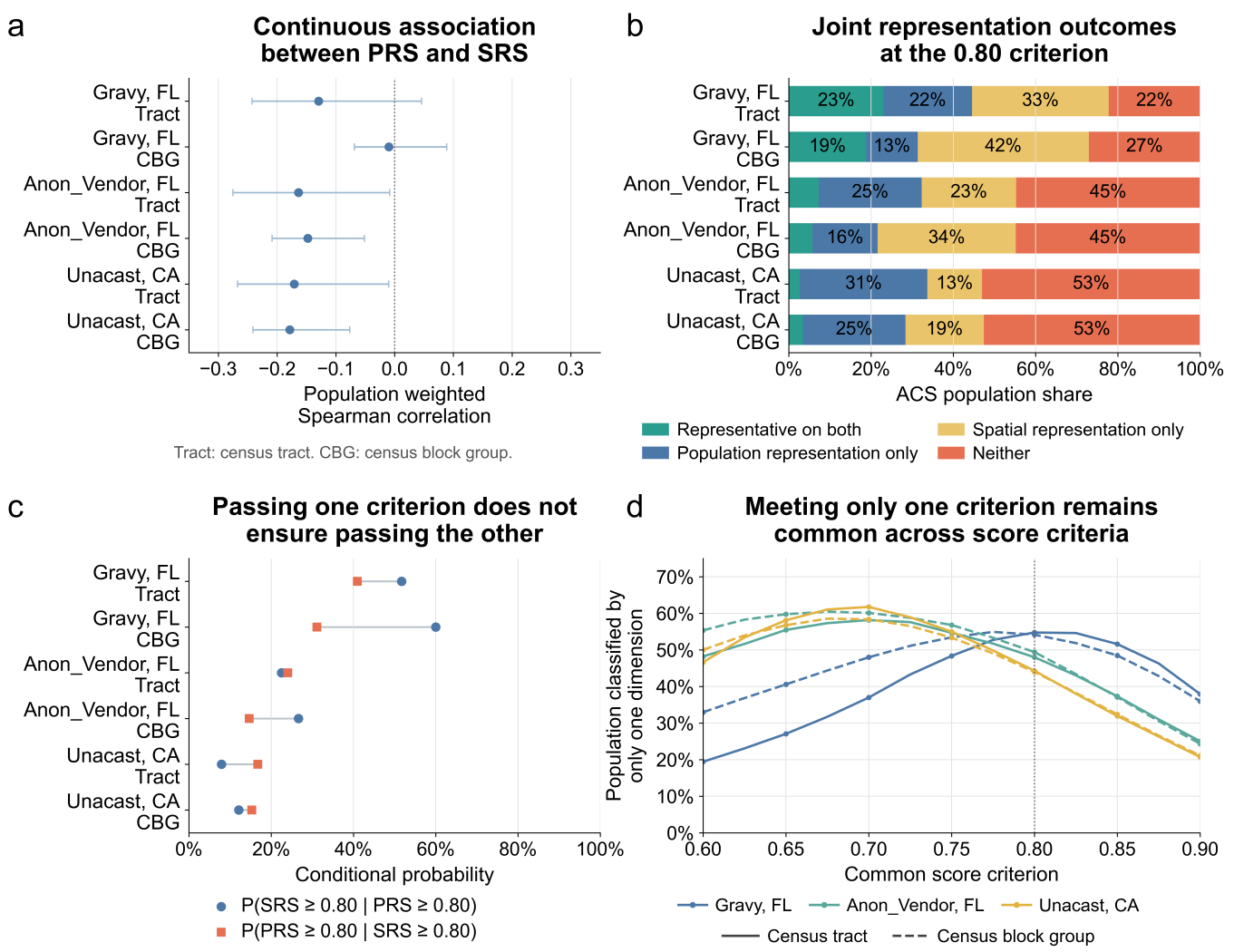


Fig. 2. Population and spatial representation describe distinct dimensions. (a) Dots show population weighted Spearman correlations between PRS and SRS. Horizontal lines show 95% confidence intervals from 500 county cluster bootstrap samples. The vertical dashed line indicates a correlation of zero. (b) Bars show the ACS population shares in four mutually exclusive categories at the common score criterion of 0.80: representative on both dimensions, population representation only, spatial representation only, and neither. Percentages are based on the ACS population in areas where both scores could be evaluated. (c) Circles show $\Pr(SRS \geq 0.80 \mid PRS \geq 0.80)$. Squares show $\Pr(PRS \geq 0.80 \mid SRS \geq 0.80)$. (d) Lines show the ACS population share that met exactly one criterion as the common criterion varied from 0.60 to 0.90. The dotted line marks the primary criterion of 0.80. Solid lines represent census tracts. Dashed lines represent census block groups.

selected counties were retained; this approach accounts for dependence among geographic units within the same county. Numerical bootstrap results and unweighted estimates are reported in SI Appendix, Table S7. Together, these findings show that PRS and SRS capture distinct dimensions of spatial bias and should not be used interchangeably.

Applying the common 0.80 criterion further showed the disagreement between the two dimensions (Fig. 2b). Across the tract and block group comparisons, only 2.7% to 23.0% of the ACS population met both criteria. In Gravy tracts, 23.0% met both criteria, 21.5% met only the PRS criterion, 33.2% met only the SRS criterion, and 22.2% met neither. In Anon_Vendor tracts, the corresponding shares were 7.3%, 25.0%, 23.0%, and 44.7%. In Unacast tracts, they were 2.7%, 31.0%, 13.3%, and 53.0%. Across all six comparisons, 44.0% to 54.8% of the population met only one of the two criteria, further indicating that meeting one representation criterion does not imply meeting the other.

Meeting one criterion did not reliably indicate that the other criterion was also met (Fig. 2c). Among the population in areas meeting the PRS criterion, 7.9% to 60.0% also met the SRS criterion; among the population in areas meeting the SRS criterion, 14.6% to 40.9% also met the PRS criterion. Population weighted Cohen's kappa ranged from −0.155 to −0.005, indicating little agreement between the two classifications, and the area under the receiver operating characteristic curve for continuous PRS predicting whether SRS met the 0.80 criterion ranged from 0.311 to 0.492 (near or below chance). Thus, higher PRS values did not reliably identify areas with better spatial representation (SI Appendix, Table S8).

This pattern was not specific to the 0.80 criterion. Across common score criteria from 0.70 to 0.90, 20.7% to 61.8% of the population met only one criterion (Fig.

2d). The share meeting only one criterion decreased at higher criteria because more areas failed both criteria, not because of a stronger agreement between PRS and SRS. Joint maps and score distributions further show where the two classifications differed (SI Appendix, Figs. S5 to S8). Bivariate Moran's I between population mismatch in one area and spatial mismatch in neighboring areas was also close to zero, ranging from −0.010 to 0.004 at fine scales. These results show that the two dimensions were weakly related both within areas and across neighboring areas.

Higher sampling rates do not ensure better representation. Increasing sample quantity did not consistently improve spatial representation. In the observed device level data, higher local sampling rates were associated with *lower* SRS at both tract and block group scales. Population weighted Spearman correlations ranged from −0.575 to −0.373 at the tract scale and from −0.510 to −0.379 at the block group scale. The block group curves in Fig. 3a show that SRS remained stable (Unacast, Anon_Vendor) or increased slightly (Gravy) over a limited range of low sampling rates, but declined across most of the observed range. Cubic spline models supported this nonlinear pattern in all six fine scale comparisons based on the Bayesian information criterion (BIC, SI Appendix, Fig. S9 and Table S9).

We further used segmented regression to identify changes in slope in the relationship between sampling rate and SRS (Fig. 3b). The fitted relationships showed clear changes in slope across datasets and scales. Before the estimated change points, slopes were positive or close to zero in several comparisons, whereas above them, every fitted slope was negative. Estimated change points were 3.47% and 3.70% for Gravy at the tract and block group scales, 6.85% and 9.32% for Anon_Vendor, and 2.64% and 2.40% for Unacast. Uncertainty around these estimates differed across datasets: the 95% bootstrap interval for Anon_Vendor tracts extended from 6.85% to 29.58%, whereas the interval for Gravy tracts extended from 3.47% to 3.64%. Segmented models improved on linear models by 18.2 to 639.9 BIC points, and a tenfold increase in sampling rate was associated with a 0.269 to 0.410 decrease in SRS. These results show that there was no common sampling threshold for better spatial representation; instead, higher sampling rates beyond the estimated change points were consistently associated with lower SRS.

To further test whether increasing the number of observations from the same sampling process was associated with better spatial representation, we conducted controlled random thinning of inferred home locations (Fig. 3c), using the complete home sample for each dataset served as the internal reference. Twenty nested sequences retained 5%, 10%, 25%, 50%, 75%, or 100% of observed homes; we recalculated SRS at each retention level and measured its population-weighted change relative to the complete sample. Further details are provided in Materials and Methods. At 5% retention, SRS was lower than the complete-sample in all datasets and at both geographic scales. SRS generally increased with the retention rate up to 25% or 50%, reaching its maximum at 25% for Gravy, 50% for Anon_Vendor, and either 25% or 50% for Unacast depending on the spatial scale, before decreasing as the retention rate approached 100%. These patterns were nonmonotonic and differed across datasets and geographic scales. Increasing sample size, then, may reduce variation when the retained sample is small, but it did not consistently improve spatial representation; the intermediate peaks reflect the highest SRS values among the tested retention levels and should not be interpreted as optimal sampling rates. Absolute values and the full thinning design are reported in SI Appendix, Figs. S10 and S11 and Table S10.

PRS cannot be directly regressed on local sampling rate because PRS itself is determined by the local sampling rate relative to the statewide rate. We therefore used multinomial resampling to vary effective sample size while holding the observed population allocation pattern fixed. At the block group scale, retaining 10% of the sample produced total variation distances 1.028 to 1.172 times those of the full sample (Fig. 3d); at 1% retention, the ratios ranged from 1.257 to 2.153. As the retained share increased, the ratios approached one. However, the full sample total variation distance remained between 0.115 and 0.315. Increasing sample size therefore reduced additional finite-sample error without removing the population allocation mismatch already present in the observed data. Results for all three scales and 200 resamples per condition are reported in SI Appendix, Figs. S12 and S13 and Table S11. Taken together, these analyses show that larger samples reduce sampling variation but do not, on their own, improve spatial representation or remove existing population allocation bias.

Associated factors differ across representation dimensions. Local demographic, socioeconomic, and built environment characteristics were associated with both forms of representation mismatch (Y_i^P and Y_i^S (see Materials and Methods for further details), but the patterns differed across outcomes and datasets (Fig. 4). Residuals from ordinary regression models showed spatial autocorrelation; therefore, we estimated spatial error models for each dataset, state, geographic scale, and available outcome. Positive coefficients indicate higher representation mismatch, whereas negative coefficients indicate lower mismatch.

Population representation mismatch showed several consistent associations across datasets (Fig. 4a,c). The strongest pooled association was with population density. Areas with greater population density had lower mismatch, with a synthesized standardized coefficient of −0.199 (95% interval: −0.262 to −0.135). Mismatch was also lower farther from the coast (−0.097, −0.125 to −0.069). In contrast, mismatch was higher in areas with more multifamily housing (0.151, 0.129 to 0.174) and a larger share of residents aged 65 or older (0.078, 0.041 to 0.115). Smaller positive associations were found for non-White population share (0.045, 0.015 to 0.076) and building density (0.044, 0.014 to 0.073). These synthesis estimates summarize 14 dataset-, state-, and scale-specific models. Several factors showed substantial between-model heterogeneity, so the pooled estimates should be interpreted as overall patterns rather than associations that were uniform across all datasets.

Spatial representation mismatch showed a different pattern (Fig. 4b,c). Mismatch was lower in areas with greater building density (−0.106, −0.123 to −0.089)

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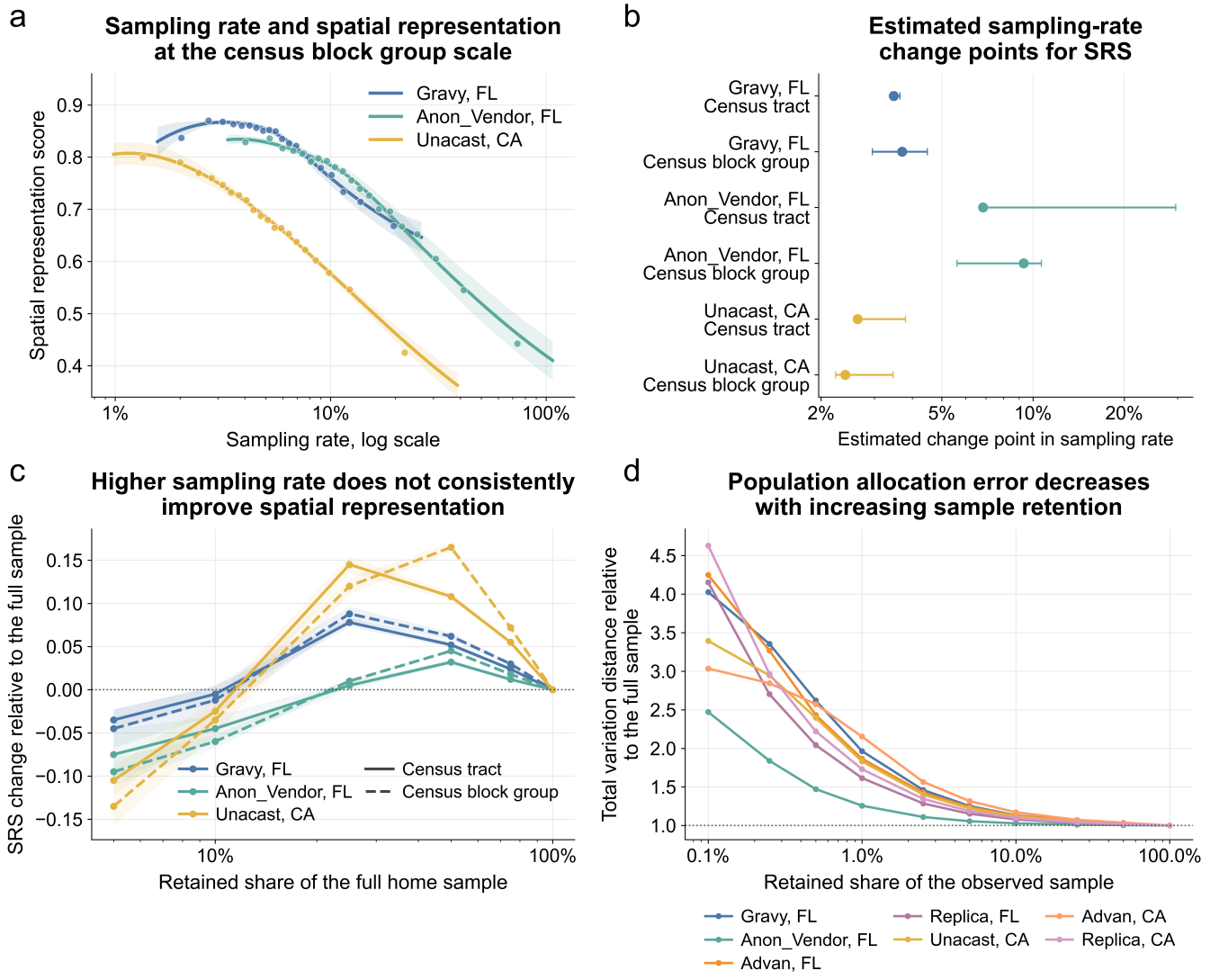


Fig. 3. Sampling quantity does not remove spatial bias. (a) Curves show fitted cubic spline relationships between sampling rate and SRS at the census block group scale. Shaded bands show 95% county cluster confidence intervals. Points show population weighted mean SRS within sampling rate bins. (b) Points show sampling rate change points estimated from segmented regression at the census tract and block group scales. Horizontal lines show 95% county cluster bootstrap intervals. (c) Controlled thinning of device level home points reports the change in population weighted mean SRS relative to the internal full sample baseline. Positive values indicate that the thinned sample has a higher SRS than the full sample. Solid and dashed lines denote census tracts and census block groups. Shaded bands are 95% county cluster intervals across 20 nested thinning sequences. (d) Multinomial resampling shows census block group total variation distance relative to the observed full sample for all seven provider and state combinations. Values above one indicate additional sample error compared with baseline. Ratios approach one as the retained share approaches 100%.

and a larger share of residents age 65 or older (−0.057, −0.080 to −0.033). It was higher in areas with a larger share of residents with a bachelor's degree (0.076, 0.053 to 0.100) and more households without internet access (0.015, 0.003 to 0.027). Population density had a positive pooled estimate, but its interval included zero. These associations differed from those observed for population representation mismatch. For example, population density was strongly negatively associated with population mismatch but showed no clear pooled association with spatial mismatch. Building density was positively associated with population mismatch but negatively associated with spatial mismatch. These contrasting patterns further show that population allocation and within-area home placement capture different dimensions of representation.

The spatial error models reduced residual spatial dependence but had little effect on the overall coefficient patterns (Fig. 4d). The spatial error parameter, λ, was positive in all 20 models, with a median of 0.289 and a range from 0.055 to 0.711. This pattern indicates residual spatial dependence that was not explained by the nine observed factors. Coefficients from the spatial error models were highly correlated with those from identically specified ordinary regression models. Spearman correlations were 0.988 for population mismatch and 0.987 for spatial mismatch. Median residual Moran's I decreased from 0.074 to −0.008 for population mismatch and from 0.073 to −0.007 for spatial mismatch. Alternative models using five or 12 nearest neighbors produced coefficient correlations above 0.997 with the primary specification. Spatial lag models and additional diagnostics produced

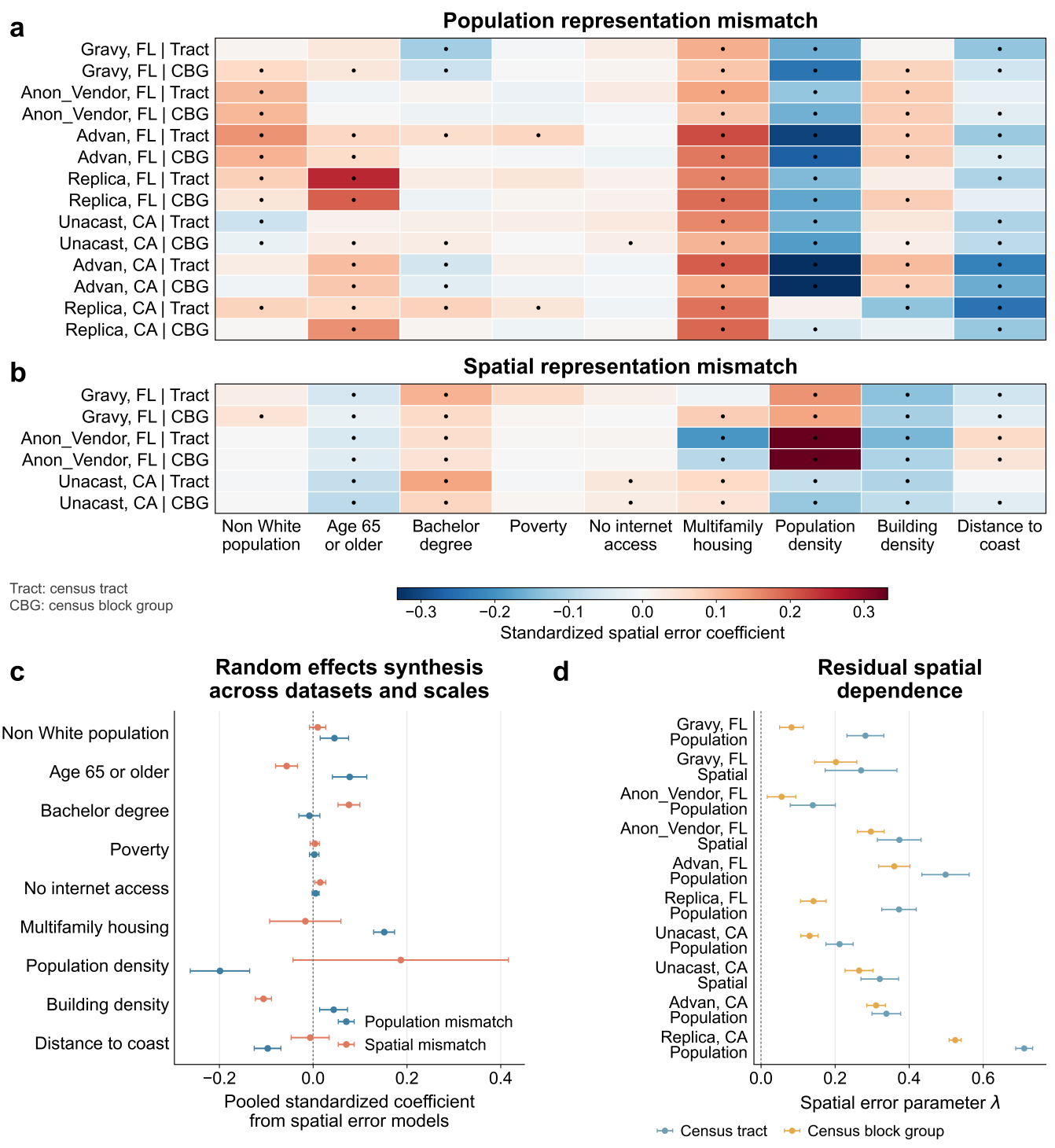


Fig. 4. Area characteristics associated with population and spatial representation mismatch. (a) Heat map of standardized coefficients from 14 spatial error models of population representation mismatch. Rows identify provider, state, and geographic scale, and columns identify the nine focal factors. (b) Corresponding standardized coefficients from six models of spatial representation mismatch. Warm colors indicate positive associations with mismatch, and cool colors indicate negative associations. Black dots indicate false discovery rate-adjusted $q < 0.05$. (c) Points and 95% intervals show random-effects synthesis estimates across dataset, state, and scale-specific models. Blue indicates population mismatch, and coral indicates spatial mismatch. (d) Points and 95% intervals show the spatial error parameter, λ, by outcome and geographic scale. All models use symmetric, row-standardized weights based on the eight nearest geographic centroids.

similar conclusions (SI Appendix, Figs. S14 to S20 and Tables S15 and S16).

Discussion

This study evaluated two dimensions of spatial representation in five mobility data products, including Gravy, Anon_Vendor, Unacast, Advan, and Replica. The

analysis covered Florida and California and included seven combinations of providers and states. We assessed population representation by comparing the geographic allocation of estimated residents with ACS population estimates at the county, census tract, and census block group scales. For Gravy, Anon_Vendor, and Unacast, which provided individual device records and inferred home locations, we also compared observed home spacing with simulated reference patterns within mapped residential land. We used resampling experiments to examine the role of sample size and spatial error models to identify local factors associated with representation mismatch. Our purpose was to determine whether a large sample ensures both proportional population allocation and the expected spatial distribution of inferred homes. Four main findings emerged. First, population representation was scale dependent, with greater mismatch at finer geographic scales. Second, population and spatial representation captured distinct dimensions. A tract could receive the expected share of estimated residents while its inferred homes were concentrated in a small part of the residential area. Conversely, its population allocation could differ from the census benchmark while its inferred homes still followed the expected spatial pattern within the tract. Third, multinomial resampling showed that larger samples reduced random variation but did not remove existing population mismatch. Controlled thinning further showed that adding inferred homes did not consistently improve spatial representation. Finally, demographic, socioeconomic, and built environment characteristics were associated with both forms of mismatch, but their associations differed between them. These findings show that spatial representation depends on scale, the outcome being evaluated, and the process through which observations are collected and assigned.

Population allocation mismatch increased from counties to census tracts and census block groups in every provider and state combination (Fig. 1c). This pattern is consistent with an aggregation effect. When overrepresented and underrepresented areas are combined, their errors can offset one another, making county totals appear closer to the population benchmark. Spatial aggregation can therefore mask variation within larger geographic units (18, 21). This is the scale effect of the modifiable areal unit problem (MAUP), in which results change when the same observations are summarized using geographic units of different sizes (18). It also supports broader concerns about evaluating large geographic datasets only at aggregated scales (19). Similar declines in agreement between mobile device counts and Census population have been reported at finer scales (16). The pattern also appeared for Replica, although its closer agreement with the ACS is partly expected because Census controls inform its synthetic population. In contrast, spatial representation did not show the same monotonic scale pattern (Fig. 1d). SRS increased modestly in some comparisons, possibly because smaller units contain more internally similar residential environments. A product that matches the Census benchmark at the county scale can therefore still allocate residents unevenly among the tracts or block groups within that county. Representation should be evaluated at the same scale as the intended analysis. Coarser units may yield lower measured population mismatch, while finer units preserve the local variation needed for neighborhood or road level analyses (22). Finer resolution does not itself imply greater accuracy. Instead, our results point to a practical tradeoff between apparent agreement at aggregated scales and the analytical detail available at finer scales.

We also found that population and spatial representation captured distinct dimensions of representation (Fig. 2). Their continuous scores were only weakly related, their classification agreement was close to zero, and substantial shares of the population met only one criterion. This difference follows from how the two measures are defined. PRS measures population allocation across geographic areas. It compares each area's share of estimated residents with its share of the ACS population. SRS measures the spatial pattern of inferred homes within an area. Conditional on the number of homes observed, it compares their spacing with reference patterns generated within mapped residential land. An area can receive the expected share of estimated residents while its inferred homes remain concentrated in only part of the residential area. Conversely, its share of estimated residents can differ from the census benchmark while its inferred homes still follow the expected internal spacing. This reflects a difference in spatial support. PRS summarizes population allocation among geographic units, whereas SRS summarizes a point pattern within each unit (23). A high SRS indicates agreement in the overall spacing pattern. It does not confirm that each inferred home is located at its true address. Studies that use inferred homes to assign environmental exposures or neighborhood characteristics should therefore evaluate both population allocation and spatial distribution.

The sampling experiments further distinguished limited sample size from persistent mismatch in the data process (Fig. 3). In multinomial resampling, increasing sample size reduced additional random variation, but the estimates converged to the full sample allocation rather than the population benchmark. Controlled thinning produced a related result for spatial representation. SRS changed nonlinearly with the retained fraction, and the pattern differed across products and scales. These findings are consistent with the big data paradox, in which a large sample can estimate a biased quantity with high precision (14). Increasing the number of records alone cannot correct systematic differences in app participation, location sharing, home inference, or vendor processing. When added records come from the same selective process, they may reinforce existing geographic coverage patterns rather than fill gaps in poorly represented areas (14, 16). For SRS, the locations and continuity of additional observations also matter. Verma et al. found that home detection performance depended more on the temporal and spatial continuity of GPS traces than on total data volume (9). This evidence supports our interpretation that trace structure and inference procedures, rather than sample size alone, shape the spatial pattern of inferred homes. The negative associations between observed sampling rate and SRS should not be interpreted as causal effects. The estimated change points describe product specific patterns and should not be treated as recommended sampling rates.

The factor models identified several local correlates of population mismatch (Fig. 4a,c). These associations may arise at different stages of the data process, including

DRAFT

smartphone access, app participation, location sharing, home inference, and vendor processing. Population density was associated with lower mismatch, but this result should not be explained by sample size alone. Previous studies show that urban and rural coverage differences vary across years, geographic scales, and data sources (16, 17). Density may instead reflect differences in app use, provider coverage, housing form, and home detection conditions. Areas with more older residents and larger racial minority populations also had greater population mismatch. Commercial mobility panels depend on selected applications and users who permit location sharing. These processes can create demographic differences in who contributes usable traces. Previous audits have found uneven demographic coverage, although the direction varies across places and periods (12, 16). Because our outcome measures the magnitude of mismatch, these results do not show that older or racial minority populations were consistently underrepresented. Population mismatch was also greater near the coast. Tourists, seasonal workers, and second home residents may spend extended periods away from their Census residence. Morrison et al. found that seasonal workers could be classified as local residents after several weeks because home was inferred from recent nighttime activity (24). Seasonal mobility has also been documented in coastal destinations and among temporary residents in Florida (25, 26). These processes could shift inferred homes toward coastal areas while the ACS continues to use usual residence. However, this pathway cannot be tested without information about temporary residents or changes in device residence.

Built environment and socioeconomic factors were associated with both population and spatial mismatch (Fig. 4b,c). Multifamily housing was associated with greater population mismatch. Apartment buildings place many households within the same horizontal footprint, which may complicate home assignment. Home estimates are also sensitive to the observation period and inference rule (8). Building density was associated with slightly greater population mismatch but lower spatial mismatch. The latter result does not indicate more accurate locations. Closely spaced buildings can block and reflect satellite signals, which can increase positional uncertainty (27). Lower spatial mismatch may instead reflect closer agreement between compact development and the residential land reference. This reference does not account for building height or housing units within each structure (28). Spatial mismatch was also greater in areas with more college educated residents and in areas with more households without internet access. These results do not suggest a single socioeconomic pattern. Home internet is not a direct measure of smartphone access or location sharing, while education may relate to the applications through which traces are collected. Existing studies mainly examine population coverage rather than the internal spacing of inferred homes (15, 29). Likewise, lower spatial mismatch in areas with more older residents does not indicate better coverage of older adults because SRS is conditional on the homes already observed. The significant spatial error parameters further indicate that unmeasured spatial processes remained (Fig. 4d). Possible sources include regional app coverage, land use, privacy filters, network conditions, and geocoding procedures. Provider comparisons have documented differences across commercial mobility sources (20). However, the spatial error parameter does not identify which process produced the residual dependence (30). These factors should therefore be treated as contextual correlates rather than causal effects. Representation should be evaluated for each product, scale, and intended application, and sampling rate should not replace a direct representation audit.

This study has several limitations. It examined two states, specific observation periods, and five data products. SRS was available for only three products. Provider collection and processing procedures differ and are reported at different levels of detail. We therefore evaluated each delivered product as a whole rather than assigning mismatch to a single processing step. The ACS provides a common external benchmark but includes estimation error. Replica also uses Census controls as part of its calibration, so its agreement with the ACS is not an independent validation. The residential reference sampled locations in proportion to an independent estimate of residential population density. This weighting captures broad differences in expected residential concentration, but it may not fully represent the distribution of households across parcels, buildings, and floors. SRS also summarizes the point pattern using the mean distance to five nearest neighbors and does not capture all forms of spatial clustering or dispersion. Finally, the factor models identify associations rather than causal effects. Future research should evaluate additional regions, seasons, and data providers. Residential references could combine population density with housing unit counts, building footprints, and building height where consistent data are available. Longitudinal residence histories could help test seasonal and coastal mechanisms. Further work should also examine how population and spatial mismatch affect exposure estimates, accessibility measures, evacuation studies, and other applications that depend on inferred home locations.

Materials and Methods

Study design and mobility data. We analyzed five commercial mobility datasets in Florida and California. Gravy and Anon_Vendor were evaluated in Florida, and Unacast was evaluated in California. These three datasets contained device level location records from which home coordinates could be inferred. Advan and Replica were evaluated in both states. Both products provided aggregate counts by home area. They supported analyses of population representativeness but not analyses of spatial representativeness at the point level. For Advan, we used the `NUMBER_DEVICES_RESIDING` field, which reports the number of distinct observed devices with a primary nighttime location in each census block group (CBG). We treated this count as a proxy for sampled residents rather than as a direct count of residents. Replica is an aggregate product derived from a synthetic population system. Across the seven combinations, dataset size was measured using home-assigned devices or synthetic residents, depending on the product. Dataset sizes ranged from approximately 1.03 million to 15.08 million. Statewide sampling rates ranged from 2.7% to 38.4%. The overall data coverage windows were September 1 through October 15, 2022, in Florida and December 1,

DRAFT

2024, through February 28, 2025, in California. For the three device level products, the home-inference window was restricted to the 14-day period before the onset of the corresponding disaster. The overall coverage windows and home-inference periods are reported separately in SI Appendix, Table S1.

Analyses were conducted at the county, census tract, and census block group levels. Population benchmarks came from the U.S. Census Bureau's American Community Survey five-year estimates. Geographic boundaries came from the TIGER/Line shapefiles. For each provider and state combination, the same population benchmark was used across all geographic scales. Detailed observation periods, counts, geographic coverage, and census vintages are reported in SI Appendix, Table S1.

Home inference and residential reference. Home locations for Gravy, Anon_Vendor, and Unacast were inferred with the Grid-based Home Detection via Stay Time method (GHOST) (10). GHOST used only observations from the pre-event 14-day baseline period; observations recorded on or after disaster onset were excluded from home inference. The resulting pre-event home estimates were used to evaluate residential representation in the three device level products. The algorithm projects device traces to a local coordinate system, finds the nighttime grid cell with the longest observed time span, and refines the location within that cell. A weekend daytime rule is used when nighttime evidence is insufficient. The same procedure was applied to all three device level sources. Advan and Replica resident totals were read from their aggregate home area fields and were not assigned point coordinates.

Residential land was identified using the Florida Statewide Land Use Land Cover dataset and the California General Plan Land Use dataset. Residential population density was derived from WorldPop Global2 constrained (31), using the 2022 and 2024 population estimates for Florida and California, respectively, at a resolution of 3 arc (approximately 100m at the equator). The reference surface was obtained independently of the commercial mobility samples. Data preparation and the construction of sampling weights are described in SI Appendix, Supplementary Methods. Florida points were analyzed in Universal Transverse Mercator Zone 17 North (EPSG:32617). California points used California Albers (EPSG:3310). Detailed land use classes and processing rules are reported in SI Appendix, Supplementary Methods.

Population representativeness. Population representativeness measures whether the spatial distribution of a mobility sample is proportional to the benchmark population distribution. It focuses on sample counts across geographic units rather than the positional accuracy of inferred home locations. The ACS population was used as the benchmark. Deviations from the benchmark distribution indicate geographic sampling bias.

Let M_i denote the sample count from a mobility dataset in geographic unit i. Let P_i denote the corresponding ACS population. For units with $P_i > 0$, the local sampling rate is

$$SR_i = \frac{M_i}{P_i}. \quad [1]$$

We define the directional population representation ratio as

$$RR_i = \frac{M_i / \sum_j M_j}{P_i / \sum_j P_j}. \quad [2]$$

At each geographic scale, the sums include all units within the state. The ratio compares the mobility sample share of an area with its share of the census population. It is also equal to the local sampling rate divided by the statewide sampling rate. A value of one indicates proportional representation. Values below one indicate underrepresentation. Values above one indicate overrepresentation.

We then define a bounded population representation score as

$$PRS_i = \min\left(RR_i, RR_i^{-1}\right) \quad [3]$$

when $RR_i > 0$. We set $PRS_i = 0$ when $M_i = 0$. The reciprocal form treats proportional underrepresentation and overrepresentation symmetrically. The score ranges from zero to one. Higher values indicate closer agreement with the benchmark population distribution. For descriptive analysis, we classified areas with $PRS_i \geq 0.80$ as having good population representativeness. This criterion is equivalent to $0.80 \leq RR_i \leq 1.25$.

We also evaluated agreement between the complete mobility and census distributions. Total variation distance was calculated as

$$TVD = \frac{1}{2} \sum_i |m_i - p_i|, \quad [4]$$

where $m_i = M_i / \sum_j M_j$ and $p_i = P_i / \sum_j P_j$. We also calculated Jensen Shannon divergence using base two logarithms (32). Finally, we calculated the Lorenz Gini coefficient of RR_i, weighted by p_i. For all three measures, a value of zero indicates perfect agreement. Lower values indicate closer agreement between the mobility and census distributions.

Spatial representativeness. Population representativeness evaluates how sample counts are allocated across geographic units. It does not assess how inferred homes are spatially distributed within each unit. A geographic unit can have a high sampling rate even when most sampled homes come from only a small part of its residential population. However, spatial clustering does not necessarily indicate bias. Areas with higher residential density are expected to contain more home locations. We therefore compared the empirical home locations with a residential reference process that accounts for variation in residential population density.

Following the home inference procedure, each point represents the inferred home of a sampled device. We treated these locations as a spatial point process (33, 34). Let C_i denote the number of inferred homes in geographic unit i. For each home, we calculated the mean distance to its five nearest neighbors. We then averaged these distances across all inferred homes in the unit. The resulting empirical average five nearest neighbor distance was denoted by r_i (35–37). Units with fewer than six inferred homes were not evaluated.

We constructed the reference distribution using Monte Carlo simulation (38). In each geographic unit, mapped residential land defined the eligible spatial support, and

DRAFT

an independently estimated residential population density surface determined the relative probability of sampling locations within that support. Each repetition generated exactly C_i reference points, with sampling probabilities normalized within the unit. This reference accounts for spatial variation in residential population density while holding the number of homes fixed at the observed count, allowing the comparison to focus on local spatial configuration conditional on sample size.

The simulation was repeated 100 times. The same average five nearest neighbor distance was calculated for each repetition. Let h_i denote the mean reference distance across all repetitions. A value of $r_i < h_i$ indicates that the inferred homes are more clustered than expected. A value of $r_i > h_i$ indicates that they are more dispersed than expected. This comparison measures agreement in local spatial spacing rather than exact overlap between empirical and reference points.

For $r_i > 0$ and $h_i > 0$, we define the spatial representativeness score as

$$SRS_i = \exp\left[-\left|\log\left(\frac{r_i}{h_i}\right)\right|\right] = \min\left(\frac{r_i}{h_i}, \frac{h_i}{r_i}\right). \quad [5]$$

The reciprocal form assigns the same score to reciprocal deviations below and above one. It therefore treats proportional clustering and dispersion symmetrically. For positive distances, SRS_i lies in $(0, 1]$. A value of one indicates exact agreement in average local spacing. Smaller values indicate greater disagreement with the reference distribution. For descriptive analysis, we classified units with $SRS_i \geq 0.80$ as having good spatial representativeness. This criterion is equivalent to $0.80 \leq r_i/h_i \leq 1.25$.

The full Monte Carlo distribution was also used to test whether the inferred homes were significantly more clustered or dispersed than the reference distribution. Simulation procedures and hypothesis testing details are provided in SI Appendix, Supplementary Methods.

Magnitude, scale, and dimensional distinction. We calculated all population representativeness metrics for seven provider and state combinations at three scales. Paired Wilcoxon signed rank tests compared scale changes across combinations. Population weighted empirical distribution functions retained the full PRS and SRS distributions. We measured spatial clustering of population mismatch, $\log(1+|RR_i-1|)$, with global Moran's I (39). The primary weights were row standardized and based on eight nearest geographic centroids. Inference used 999 permutations. Alternative nearest neighbor and Queen weights were tested in the SI Appendix.

The dimensional analysis was limited to units with both PRS and SRS. We calculated population weighted Spearman correlations with 500 county cluster bootstrap samples. At a common score criterion, each area's population was classified as representative on both dimensions, PRS only, SRS only, or neither. We also calculated both conditional probabilities, population weighted Cohen's kappa, and population weighted area under the receiver operating characteristic curve. Criterion sensitivity covered 0.50 to 0.95. Bivariate Moran's I tested whether population mismatch in one area was related to spatial mismatch in neighboring areas.

Sampling intensity experiments. Cross sectional SRS models used $\log_{10}(SR_i)$. We reported population weighted and unweighted Spearman correlations. Cubic spline models used four through eight degrees of freedom. The model with the lowest BIC was selected, with standard errors clustered by county (40). Segmented regression used a continuous linear spline in log sampling rate (41). Candidate change points covered the 10th through 90th percentiles. We selected the value with the lowest BIC and required an improvement of at least six over a linear model. Uncertainty came from 250 county cluster bootstrap samples.

Controlled thinning (42) used the raw inferred home point files. Areas required at least 200 homes in the full point file. Twenty nested random sequences retained 5%, 10%, 25%, 50%, 75%, and 100% of homes. Empirical five nearest neighbor distances were recalculated. Residential reference distances were scaled for the retained point intensity according to the inverse square root relation for planar nearest neighbor distance. Each sequence was compared with its own 100% point file baseline. This internal baseline is separate from the harmonized stored SRS used in the cross sectional analysis.

For population allocation, we drew 200 multinomial samples at effective sample shares from 0.1% to 100%. Draw probabilities were fixed at each product's observed geographic population shares. This design changes sample size while holding the observed allocation process constant. Total variation distance was divided by its observed full sample value. Ratios above one measure additional finite sample error. Convergence to one means convergence to the observed full sample mismatch, not to the census distribution.

Factors associated with representation mismatch. The population representation outcome variables were

$$Y_i^P = \log\left(1 + |RR_i - 1|\right). \quad [6]$$

The spatial outcome was

$$Y_i^S = -\log(SRS_i). \quad [7]$$

Higher values indicate greater mismatch. Population models included all seven provider and state combinations at tract and block group scales. Spatial models included the three device level combinations at both scales.

Nine predictors measured non White population share, share age 65 or older, bachelor degree share, poverty share, lack of household internet access, multifamily housing share, population density, building density, and distance to the coast. Population density, building density, and distance were transformed as $\log(1 + x)$. Outcomes and covariates were standardized within each provider, state, and scale sample. Spatial mismatch models also controlled for standardized $\log(1 + 100SR_i)$. Population mismatch models did not control sampling rate because RR_i is defined by relative sampling rate. Missingness was below 0.37%, and the maximum variance inflation factor was 3.37 (SI Appendix, Table S12).

The primary specification was a spatial error model,

$$Y = X\beta + u, \qquad u = \lambda W u + \epsilon. \quad [8]$$

DRAFT

Models were estimated using heteroskedastic generalized method of moments estimation (30). All geographic units received equal observation weight. The spatial weights matrix W was constructed from the symmetric union of each area's eight nearest geographic centroid neighbors. Neighbors were identified using great circle distance, and the resulting matrix was row standardized. Cubic spline bases for longitude and latitude, each with four degrees of freedom, controlled for broad spatial trends. The Benjamini-Hochberg procedure was applied across the nine focal coefficients within each model (43).

For each outcome and predictor, we synthesized standardized coefficients from the primary spatial error models across datasets and spatial scales. The population representation synthesis combined 14 model estimates, and the spatial representation synthesis combined six estimates. Let $\hat{\beta}_m$ and s_m denote the standardized coefficient and its standard error from model m. Random effects weights were defined as

$$w_m = \frac{1}{s_m^2 + \tau^2}, \qquad [9]$$

where τ^2 is the between model variance estimated using the DerSimonian and Laird method (44). The pooled coefficient was $\hat{\beta}_{\text{pooled}} = \frac{\sum_m w_m \hat{\beta}_m}{\sum_m w_m}$, with standard error $\left(\sum_m w_m\right)^{-1/2}$. Confidence intervals were calculated as $\hat{\beta}_{\text{pooled}} \pm 1.96SE$. This approach accounted for both coefficient uncertainty within each model and heterogeneity across datasets and spatial scales. Benjamini Hochberg adjustment was also applied across the nine pooled coefficients within each outcome. We report τ^2, I^2, coefficient sign consistency, and the complete dataset and scale specific estimates in the SI Appendix.

Spatial lag models, ordinary regression models, residual Moran tests with 99 permutations, and spatial weights based on five and 12 nearest neighbors were used as sensitivity analyses. Full model equations, synthesis diagnostics, and sensitivity results are provided in SI Appendix, Supplementary Methods.

Data, Materials, and Software Availability. Analysis code and derived geographic unit metrics will be deposited in a public repository before publication. Raw commercial mobility records cannot be redistributed because of data license restrictions. They can be made available to editors and reviewers when permitted by the relevant agreements. American Community Survey and geographic boundary data are publicly available from the US Census Bureau.

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DRAFT

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DRAFT